



Negotiation Copilot: Workflow and Interface Report

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Link: https://github.com/JingjunXu/negotiation_tool

Project Overview

Negotiation Copilot is a local web application designed to address three practical challenges in negotiation preparation and execution: (1) relevant facts and constraints are dispersed across multiple documents; (2) negotiators may lose track of their redlines and walk-away conditions during live discussions; and (3) generating creative, interest-based options under time pressure is difficult. To address these challenges, the application combines multi-format ingestion with **LLM-assisted extraction** to organize source materials and **summarize supporting evidence**. It then applies **interest analysis and option generation** to help users identify priorities, formulate potential trades and packages, and refine their redlines and bottom lines. These outputs are presented through a **human-in-the-loop Streamlit interface** that enables users to verify evidence, revise model-generated content, confirm critical limits, and retain control over strategic decisions.

Workflow

The workflow converts source materials into a reviewable case, strategic analysis, and live negotiation support.



Implementation note. Automated tests primarily use offline or simulated clients; browser-level testing with live LLM API calls remains limited. Current API-integration issues have prevented complete validation of all interface interactions, so some API-dependent functions are not yet fully implemented or stable.

Future Adaptation

Future work should evaluate whether the web interface operates reliably under valid API calls. It should also streamline information presentation to reduce cognitive overload and improve interpretability. Further development could focus on a live, stage-sensitive interface that provides context-dependent strategic recommendations throughout different phases of the negotiation.

Representative Interfaces

Materials

Materials

Upload or paste negotiation materials and set a scope for each one.

Showing a demo case. Upload or paste your own materials below to start a real one.

Upload files

PDF, DOCX, TXT, or Markdown

01_News_Source.pdf

01_News_Source.pdf

01_News_Source.pdf

Scope for 01_News_Article_en_German_Ult_Source.pdf

Shared with counterpart

Scope for 01_News_Article_en_German_German_Source.pdf

Shared with counterpart

Scope for 01_News_Article_en_Beta_Swiss_Source.pdf

Shared with counterpart

Process uploaded files

Negotiation Map

Negotiation Map

Notes, context, objectives, issues, constraints, deadlines, conflicts, and open questions.

My role	Counterpart role	Context	Objective	BATNA
Vendor negotiating renewal	Client procurement lead	Annual contract renewal	Secure renewal above information value	Secondary buyer at 10% discount
Explicit	Explicit	Explicit	Explicit	Explicit
Evidence (3)	Evidence (3)	Evidence (3)	Evidence (2)	Evidence (2)
CL	DR	DR	DR	DR

Interests (raw, pre-analysis)

- Fast payment

Evidence (2)

Positions

- \$120k in 30 days

Evidence (2)

Strategy Lab

Strategy Lab

Interests

Agreement Landscape

Walk-Away

Limits

Options & Packages

Run interest analysis

My interests

- [need] Need cash flow within 30 days

Their interests (hypotheses)

- [fear] Hypothesis — verify in conversation: Fear of setting a bad precedent inferred

Verify: Would a one-time exception concern you for future deals?

Positions ↔ interests

- (me) We want \$120k in 30 days → ['Need cash flow within 30 days']

Reframe: What would faster payment let you do?

Brief Builder

Brief Builder & Export

Generate Prep Brief

No brief yet. Click 'Generate Prep Brief'.

Figure 1. Representative interfaces for material ingestion, case mapping, strategy analysis, and brief generation.